

Linear Algebra

Chapter 2 - Matrices, Determinants, System of linear equations

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Overview

- 1 Basic concepts of matrices
- 2 Matrix operations
- 3 Determinants
- 4 Rank of a matrix, inverse of a matrix
- 5 Systems of linear equations



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Definition

A matrix is a rectangular array of numbers, symbols, or expressions, arranged in rows and columns.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

The numbers, symbols, or expressions in the matrix are called its **entries** or its **elements**.

The horizontal and vertical lines of entries in a matrix are called rows and columns, respectively.

The entry (element) in the ***i*-th row** and ***j*-th column** of a matrix A is denoted by ***a_{ij}***

We denote by $A = [a_{ij}]_{m \times n}$

Our lectures focus on real and complex matrices, that is, matrices whose elements are real numbers or complex numbers, respectively.

Examples

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} i & 1+i \\ 2 & 4 \\ 1 & 2 \end{bmatrix}$$



Definition

- i) The size of a matrix is defined by the number of rows and columns that it contains. A matrix with m rows and n columns is called an $m \times n$ matrix or m -by- n matrix.
- ii) Matrices with a single row are called row vectors, and those with a single column are called column vectors.
- iii) A matrix with the same number of rows and columns ($m = n$) is called a **square matrix**.



Matrices

Name	Size	Examples
Row vector	$1 \times n$	$\begin{bmatrix} 3 & -1 & 8 \end{bmatrix}$
Column vector	$m \times 1$	$\begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}$
Square matrix	$n \times n$	$\begin{bmatrix} 1 & -2 & 3 \\ 4 & 8 & 8 \\ 5 & 3 & 7 \end{bmatrix}$



Definition

Let $A = [a_{ij}]_{nn}$ be a square matrix

- i) The elements $a_{11}, \dots, a_{nn}$ form the **main diagonal** of a square matrix. They lie on the imaginary line that runs from the top left corner to the bottom right corner of the matrix.
- ii) If all elements of A below (above) the main diagonal are zero, A is called an **upper (lower) triangular matrix**.
- iii) If $a_{ij} = 0, \forall i \neq j$, then A is called a diagonal matrix.



Matrices

Name	Examples with $n = 3$
Diagonal	$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$
Lower triangular matrix	$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$
Upper triangular matrix	$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$



Definition

The **identity matrix** I_n of size n is the n -by- n matrix in which all the elements on the main diagonal are equal to 1 ($a_{ii} = 1$) and all other elements are equal to 0 ($a_{ij} = 0, \forall i \neq j$).

$$I_1 = [1], I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \dots, I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$



Definition

A **zero matrix** or **null matrix**, denoted by 0 , is a matrix all of whose entries are zero.

$$0_{1,1} = [0], \quad 0_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad 0_{2,3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Equality of Matrices

Definition

We say that **two matrices are equal** if they have the same size and their entries are equal.

$$A = [a_{ij}]_{m \times n} = B[b_{ij}]_{m \times n} \Leftrightarrow a_{ij} = b_{ij}, \forall i, j$$

Consider three matrices shown below

$$A = \begin{bmatrix} 1 & x \\ y & -3 \end{bmatrix}, B = \begin{bmatrix} a & 7 \\ -8 & b \end{bmatrix}, C = \begin{bmatrix} 1 & 7 & z \\ m & n & p \end{bmatrix}$$

$$A = B \Leftrightarrow \begin{cases} a = 1 \\ b = -3 \\ x = 7 \\ y = -8 \end{cases} \quad C \neq A \text{ and } C \neq B$$



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Matrix addition

Definition

Let A, B be two matrices of the same size. The **sum of A and B** , denoted by $A + B$, is computed by adding the corresponding elements of A and B .

$$[a_{ij}]_{m \times n} + [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n}$$

Examples

$$\begin{bmatrix} 1 & 2 \\ 3 & -3 \\ 4 & -4 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 5 & 0 \\ 7 & 0 \end{bmatrix}$$



Scalar multiplication

Definition

Let r be a scalar and A be a matrix. The **scalar multiple** rA is the matrix whose entries are r times the corresponding entries in A

$$(-1)A = -A \Leftrightarrow A - B = A + (-1)B$$

Examples

$$2 \begin{bmatrix} 1 & 2 \\ 3 & -3 \\ 4 & -4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & -6 \\ 8 & -8 \end{bmatrix};$$

$$\begin{bmatrix} 2 & 3 & -5 \\ -1 & 0 & -6 \end{bmatrix} - 2 \begin{bmatrix} 1 & 2 & -4 \\ -3 & -2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 3 \\ 5 & 4 & -6 \end{bmatrix}$$



Properties

Let A, B and C be three matrices of the same size, and r, s be scalars.

a. $A + B = B + A$

d. $r(A + B) = rA + rB$

b. $(A + B) + C = A + (B + C)$

e. $(r + s)A = rA + sA$

c. $A + 0 = A$

f. $r(sA) = (rs)A$



Matrix multiplication

Matrix multiplication

Let $A = [a_{ij}]_{m \times p}$ be an $m \times p$ matrix and $B = [b_{jk}]_{p \times n}$ be an $p \times n$ matrix. The matrix product $C = AB$ is an $m \times n$ matrix defined by $C = [c_{ik}]_{m \times n}$ such that

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \dots + a_{ip}b_{pk}, \forall i = \overline{1, m}; k = \overline{1, n}$$

$$\begin{bmatrix} \dots & \dots & \dots \\ a_{i1} & \dots & a_{ip} \\ \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} \dots & b_{1k} & \dots \\ \dots & \dots & \dots \\ \dots & b_{pk} & \dots \end{bmatrix} = \begin{bmatrix} \dots & \dots & \dots \\ \dots & c_{ik} & \dots \\ \dots & \dots & \dots \end{bmatrix}$$

Examples

$$\begin{bmatrix} 1 & -2 \\ 3 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1.1 + (-2).3 & 1.2 + (-2).4 \\ 3.1 + (-1).3 & 3.2 + (-1).4 \\ 2.1 + (-3).3 & 2.2 + (-3).4 \end{bmatrix} = \begin{bmatrix} -5 & -6 \\ 0 & 2 \\ -7 & -8 \end{bmatrix}$$



Properties of matrix multiplication

Properties

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined

- a) $A(BC) = (AB)C$ (associative law of multiplication)
- b) $A(B + C) = AB + AC$ (left distributive law)
- c) $(B + C)A = BA + CA$ (right distributive law)
- d) $r(AB) = (rA)B = A(rB)$ (for any scalar)
- e) $I_n A = A = A I_n$ (identity for matrix multiplication)

Note that $AB \neq BA$



Power and polynomial of a matrix

Power of a matrix

Let A be a square matrix and k be a positive integer. Now, A^k denotes the product of k copies of A

$$A^k = A \dots A$$

The polynomial of a matrix

Let A be a square matrix and I be an identity matrix of the same size. If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, then

$$f(A) = a_n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I$$

Example

Find $f(A)$, where $f(x) = x^2 + 3x - 2$ and $A = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$



Definition

Given an $m \times n$ matrix A , the **transpose of A** is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

Examples: Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}, C = \begin{bmatrix} 2 & 4 & 1 & 2 \\ 5 & 0 & -1 & -3 \end{bmatrix}$$

Then

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & -3 \end{bmatrix}, B^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}, C^T = \begin{bmatrix} 2 & 5 \\ 4 & 0 \\ 1 & -1 \\ 2 & -3 \end{bmatrix}$$



Properties of the transpose of a matrix

Let A and B denote matrices whose sizes are appropriate for the following sums and products

- a) $(A^T)^T = A$
- b) $(A + B)^T = A^T + B^T$
- c) For any scalar r , $(rA)^T = rA^T$
- d) $(AB)^T = B^T A^T$



Symmetric and antisymmetric matrices

Definition

Let A be a square matrix.

- i) A is called symmetric if $A^T = A$
- ii) A is called anti-symmetric or skew-symmetric if $A^T = -A$



Basic elementary operations on matrices

Definition

Let A be a matrix. The following operations are called elementary operations:

- 1 Interchanging two rows (or two columns);
- 2 Multiplying a row (or a column) by a nonzero scalar;
- 3 Adding to one row (or one column) a multiple of another row (or column).

Notation:

- h_i denotes row i , c_j denotes column j ;
- $h_i \leftrightarrow h_j$ ($c_i \leftrightarrow c_j$): interchange row i and row j (resp. column i and column j);
- $\lambda h_i \rightarrow h_i$ ($\lambda c_i \rightarrow c_i$): multiply row i (resp. column i) by $\lambda \neq 0$ and replace row k (resp. column k);
- $h_k + \lambda h_i \rightarrow h_k$ (resp. $c_k + \lambda c_i \rightarrow c_k$): multiply row i (resp. column i) by λ and then add to row h_k (resp. column c_k).



Basic elementary operations on matrices

Example:

$$\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{h_1 \leftrightarrow h_2} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{2h_3 \rightarrow h_3} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 14 & 16 & 18 \end{bmatrix}$$

$$\xrightarrow{h_2 + (-4)h_1 \rightarrow h_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 14 & 16 & 18 \end{bmatrix}.$$



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Submatrix

For any square matrix A of size $n \geq 2$, let M_{ij} denote the (i, j) -submatrix of size $n - 1$ formed by deleting the i -th row and j -th column of A .

Examples

Given matrix $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. Find all the submatrices M_{ij} of A



Cofactor

Given $A = [a_{ij}]_n$ be a square. The (i, j) -cofactor of element a_{ij} is the number A_{ij} given by

$$A_{ij} = (-1)^{i+j} \det M_{ij}$$

Examples

Given matrix $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. Find all the cofactors A_{ij} of A .



Determinant

Definition

The determinant of a matrix A , denoted by $|A|$ or $\det A$, is defined as follows

i) 1-square matrix $|A| = |a| = a$,

ii) 2-square matrix $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

iii) 3-square matrix

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

iii) n -square matrix, $n \geq 3$

$$\det A = a_{11}A_{11} + a_{12}A_{12} + \dots + a_{1n}A_{1n},$$

where A_{ij} the cofactor of a_{ij} .

Determinant

Examples

Compute the determinant of the following matrices

$$A = \begin{vmatrix} 4 & -3 \\ -2 & -4 \end{vmatrix}, B = \begin{vmatrix} 1 & -1 & -2 \\ 2 & -3 & 1 \\ 3 & -4 & 2 \end{vmatrix}$$



Properties of the determinant

Properties

Let A, B be square matrices

- 1) $\det A = \det A^T$, i.e. Column operations have the same effects on the determinant as row operations.
- 2) The Laplace formula:

$$\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} A_{ij} = \sum_{i=1}^n (-1)^{i+j} a_{ij} A_{ij}$$

- 3) $\det(AB) = \det A \cdot \det B$
- 4) If A is a triangular matrix, then $\det A = a_{11} a_{22} \dots a_{nn}$.
- 5) If all elements of any row (column) in A are equal to zero, then its determinant is zero.

Properties of the determinant

Properties

- 6) If any two rows (columns) in A are identical (or proportional), then its determinant is zero.
- 7) If a multiple of one row (column) of A is added to another row (column) to produce a matrix B , then $\det B = \det A$.
- 8) If two rows (columns) of A are interchanged to produce B , then $\det B = -\det A$.
- 9) If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.
- 10) $\det(cA) = c^n \det A$, where n is the size of matrix A .
- 11) If elements of a row (column) in a determinant can be expressed as the sum of two or more elements, then the given determinant can be expressed as the sum of two or more determinants.



Properties of the determinant

Elementary row (column) operations

There are three kinds of elementary matrix operations.

- i) Interchange two rows (or columns).
- ii) Multiply each element in a row (or column) by a non-zero number.
- iii) Multiply a row (or column) by a non-zero number and add the result to another row (or column).

Remarks

The properties 4); 7); 8) allow to use the elementary row operations to deduce the matrix to the upper triangular matrix.

Example

Find the determinant of the matrix $A = \begin{bmatrix} -1 & 1 & 2 & 0 \\ 3 & -1 & 0 & 4 \\ -2 & 0 & 5 & -2 \\ 0 & 3 & 6 & -1 \end{bmatrix}$



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Submatrix

Definition

Let A be a matrix of size $m \times n$ and $k \leq \min\{m, n\}$. A **submatrix of size k** of matrix A , which is a square matrix of size k is obtained from A by deleting $m - k$ rows and $n - k$ columns.

Rank of matrix

The **rank of matrix A** , denoted by $r(A)$ or $\text{rank}(A)$, is the largest size of submatrix with non-zero determinant.

Remarks

If A is a matrix with $r(A) = k$, then

- i) A has at least one submatrix of size k with non-zero determinant.
- ii) The determinant of all submatrices whose size is larger than k is zero.



Properties of the rank

Properties

- i) $r(0_{m \times n}) = 0$
- ii) $0 \leq r(A) \leq \min\{m, n\}$
- iii) Let A be a square matrix of size n . Hence,
 - a) If $\det A \neq 0$, then $r(A) = n$.
 - b) If $\det A = 0$, then $r(A) < n$.



The rank of a matrix

Row echelon form

A matrix is in row echelon form if:

- ① all non-zero rows are above any rows of all zeros
- ② Each leading element of a row is in a column to the right of the leading element of the row above it
- ③ all elements in a column below a leading element are zeros

Theorem

The rank of a matrix, in a row **echelon form**, is the number of non-zero rows.

Remark

- ① Any matrix can be transformed to row echelon form by using elementary row operations.
- ② The elementary row operations on a matrix will not change its rank.

The rank of a matrix

Examples

1. Find the rank of matrix $A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & -1 & 3 \\ -4 & 5 & 2 & -1 \\ -1 & 7 & 3 & 2 \end{bmatrix}$
2. Determine the rank of the following matrix which depends on m

$$\begin{bmatrix} 1 & 2 & -2 \\ 2 & m & 1 \\ -1 & 4 & 5 \end{bmatrix}$$



Invertible matrix

Definition

Let A be a square matrix of size n . Matrix A is said to be **invertible (or non-singular)** if there exists a square matrix B of the same size such that

$$AB = BA = I_n,$$

where I_n is the identity matrix of size n . Now, B is called the inverse of A and denoted by A^{-1} .

Remarks:

- i) $(I_n)^{-1} = I_n$
- ii) The null matrix is not invertible.



Invertible matrix

Uniqueness

If A is an invertible matrix, then the inverse of matrix A , say A^{-1} , is unique.

Inverse matrix

Let A be a square matrix. If $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} C^T$$

where $C = [c_{ij}]_n$ and $c_{ij} = A_{ij} = (-1)^{i+j} \det M_{ij}$.

Examples

Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 2 \end{bmatrix}$



Inverse of a matrix

Properties

- 1) $(A^{-1})^{-1} = A$
- 2) $(kA)^{-1} = \frac{1}{k}A^{-1}$
- 3) $(AB)^{-1} = B^{-1}A^{-1}$



Inverse of a matrix

Gauss-Jordan elimination

- 1) Adjoin the identity matrix to the right side of A .
- 2) Apply row operations to this matrix until the left side is reduced to the identity matrix.
- 3) If A is reduced to I , then I becomes A^{-1} .

Using elementary operations (Gauss-Jordan method).

- ▶ **Step 1:** $[A|I] \xrightarrow{\text{Elementary operations}} [I|A^*]$.
- ▶ **Step 2:** $A^{-1} = A^*$.

Applying Gauss-Jordan elimination $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 2 & 2 \end{bmatrix}$



Inverse of a matrix

Inverse of a matrix

Let A be a square matrix with $\det A \neq 0$. Find matrix X such that

- 1) $AX = B \Rightarrow X = A^{-1}B$
- 2) $XA = B \Rightarrow X = BA^{-1}$
- 3) $AXB = C \Rightarrow X = A^{-1}CB^{-1}$
- 4) $AX + kB = C \Rightarrow X = A^{-1}(C - kB)$

Examples

Find X such that

$$X \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + 2 \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 0 & 5 \end{bmatrix}$$



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System of linear equations

Definition

A general system of m linear equations with n unknowns can be written as

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots a_{2n}x_n = b_2 \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots a_{mn}x_n = b_m \end{cases}$$

where $x_1, x_2, \dots, x_n$ are **unknowns**, a_{ij} , $\forall i = \overline{1, m}, j = \overline{1, n}$ are the **coefficients** of the system, and $b_1, b_2, \dots, b_m$ are the constant terms.



The matrix equation

The matrix equation

The system of linear equations can be written in the following form

$$Ax = b$$

where $x = [x_1, x_2, \dots, x_n]^T$, $A = [a_{ij}]_{m \times n}$ is **coefficient matrix** and $b = [b_1, b_2, \dots, b_m]^T$.

Example

$$\begin{cases} x_1 + x_2 - 2x_3 = 6 \\ 2x_1 + 3x_2 - 7x_3 = 16 \\ 5x_1 + 2x_2 + x_3 = 16 \\ x_1 + x_2 - 2x_3 = -4 \end{cases} \Rightarrow A = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 3 & -7 \\ 5 & 2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$



Gauss-Jordan elimination

Gauss-Jordan elimination

Gauss-Jordan elimination is an algorithm for solving systems of linear equations by applying **elementary operations by row**.

Let $Ax = b$ be a matrix equation. The **augmented matrix**, denoted by $\bar{A}$, is obtained by adding matrix b to the right side of A . Now, $\bar{A} = [A|b]$.

Some elementary operations by row

- Interchange two equations $\Leftrightarrow$ Interchange two rows
- Multiply one equation with $k \neq 0 \Leftrightarrow$ Multiply one row with $k \neq 0$
- Multiply one equation with k and add it to another equation $\Leftrightarrow$ Multiply one row with k and add it to another row.



Applying Gauss elimination to solve the linear system

$\overline{A} = [A|B] \xrightarrow{\text{Elementary operations}} \text{Echelon matrix } A^*, \text{ i.e. } A^* \text{ is written by}$

$$A^* = \left[\begin{array}{cccccc|c} a'_{11} & a'_{12} & \cdots & a'_{1r} & \cdots & \cdots & a'_{1n} & b'_1 \\ 0 & a'_{22} & \cdots & a'_{2r} & \cdots & \cdots & a'_{2n} & b'_2 \\ \vdots & & \ddots & \vdots & & & \vdots & \vdots \\ 0 & 0 & \cdots & a'_{rr} & \cdots & \cdots & a'_{rn} & b'_r \\ 0 & 0 & \cdots & 0 & \cdots & \cdots & 0 & b'_{r+1} \\ \vdots & & & \vdots & & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & \cdots & 0 & b'_m \end{array} \right]$$

where $a'_{kk} \neq 0$ for any $k = 1, 2, \dots, r$.

- ▶ If there exists $b'_k \neq 0$ with $k = r + 1, \dots, n$, then the system has **no solution**.
- ▶ If $b'_k = 0$ for any $k = r + 1, \dots, n$ and $r = n$, then the system has **a unique solution**.
- ▶ If $b'_k = 0$ for any $k = r + 1, \dots, n$ and $r < n$, then the system has **infinitely many solutions**.

Kronecker-Capelli's Theorem

Example: Given the system of linear equations

$$\begin{cases} x_1 + 2x_2 + 3x_3 - x_4 = 4, \\ 2x_1 + 3x_2 - x_3 = 1, \\ 3x_1 + 5x_2 + 2x_3 - x_4 = 6. \end{cases}$$

Applying Gauss-Jordan elimination on this system.

Kronecker-Capelli's Theorem

Let $Ax = b$ be a matrix equation of n unknowns (system of linear equations). The system has solution if and only if $r(\bar{A}) = r(A)$.

- 1 If $r(\bar{A}) \neq r(A)$, then the system has **no solution**.
- 2 If $r(\bar{A}) = r(A) = n$, then the system has **a unique solution**.
- 3 If $r(\bar{A}) = r(A) = r < n$, then the system has **infinitely many solutions**.



Cramer's rule

Cramer's rule

Consider a system of n linear equations for n unknowns, represented in matrix multiplication form as follows:

$$Ax = b$$

where the $n \times n$ matrix A has a non-zero determinant, and the vector $x = (x_1, \dots, x_n)^T$ is the column vector of the variables.

In this case, the system has a unique solution, whose individual values for the unknowns are given by

$$x_i = \frac{\det A_i}{\det A} \quad i = 1, \dots, n$$

where A_i is the matrix formed by replacing the i -th column of A by the column vector b .

Cramer's rule

Example

Solve the following equation

$$\begin{cases} x_1 + x_2 - 2x_3 = 6 \\ 2x_1 + 3x_2 - 7x_3 = 16 \\ 5x_1 + 2x_2 + x_3 = 16 \end{cases}$$



Homogeneous systems

Definition

A system of linear equations is homogeneous if all of the constant terms are zero:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0 \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0 \end{cases}$$

or in matrix form $Ax = 0$



Homogeneous systems

Properties

- ① Every homogeneous system has at least one solution, known as the zero solution (or trivial solution), which is $x = 0$, i.e.,
 $x_1 = x_2 = \dots = x_n = 0$.
- ② $m=n$, if $\det(A) \neq 0$, then it is also the unique solution.
- ③ $m=n$, if $\det(A) = 0$, then there are infinitely many solutions.
Moreover:
 - i) if u, v are solutions, then $u + v$ is also a solution.
 - ii) if v is a solution and k is a scalar, then kv is also a solution.



The end

Thanks for your attention!